# Stockbot / Capital Ledger — Architecture

**A single-user automated equity trading system for acting on SPTulsian advisory calls.**

Version as of 2026-08-16 · Ubuntu 22.04 on OCI · Python 3.10 · Oracle Autonomous Database

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## 1. What this system is

SPTulsian is a paid Indian equity advisory. It emails stock recommendations on market mornings and publishes targets on a member portal. Acting on those calls by hand means reading email at 9am, deciding position size against a budget policy, placing buy orders, then remembering to place a sell trigger at the advisory's target — for every open position, every day.

This system does that work unattended, and keeps a ledger of what it did.

It is deliberately **not** a trading strategy. It does not decide *what* to buy; SPTulsian does. The system decides *how much*, subject to a written budget policy, executes reliably, and reports honestly on the outcome. A separate conviction layer assesses how well each recommendation is supported by public evidence, but that layer is advisory only and cannot affect an order.

### Design priorities, in order

1. **Never place a wrong trade.** Ambiguity halts and asks a human.
2. **Never lose the ledger.** What happened must be recoverable and truthful.
3. **Fail loudly.** Silence must not be mistakable for a quiet day.
4. **Then, automate.**

Priority 1 is why the buy flow refuses to guess a ticker symbol. Priority 3 is why a scraper outage raises an alarm rather than simply producing no targets.

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## 2. Functional view

### 2.1 The daily cycle

```
09:30 IST   RECOMMEND    Read overnight advisory email.
                         Resolve each tip to a Kite trading symbol.
                         Scrape the portal for target / timeframe.
                         Write PENDING_BUY (clean) or NEEDS_REVIEW (ambiguous).
                         No money moves.
                                   |
                         90-minute review window
                                   |
10:15 IST   CONVICTION   Score each new recommendation on public evidence.
                         Display only — cannot block or size a buy.
                                   |
10:45 IST   WATCHDOG     Confirm the scraper actually ran. Email if not.
                                   |
11:00 IST   BUY          For each PENDING_BUY: price it, size it against the
                         budget policy, place a real order.
                         Re-attempt any NEEDS_REVIEW fixed in the window.
                                   |
16:00 IST   GTT          For filled buys with a target: place a GTT sell.
                         For triggered GTTs: confirm the sell actually filled,
                         then close the trade and recycle its budget.
```

The gap between 09:30 and 11:00 is the most important design decision in the system. Symbol resolution is where a catastrophic error lives — buying `ICDSLTD` (a ₹52 crore shell) when the advisory said `CDSL` (a ₹28,000 crore depository) is a real incident this system has had. Splitting recommendation from purchase creates a window in which a human can correct a bad symbol *before* money moves, rather than discovering it afterwards.

### 2.2 Position sizing

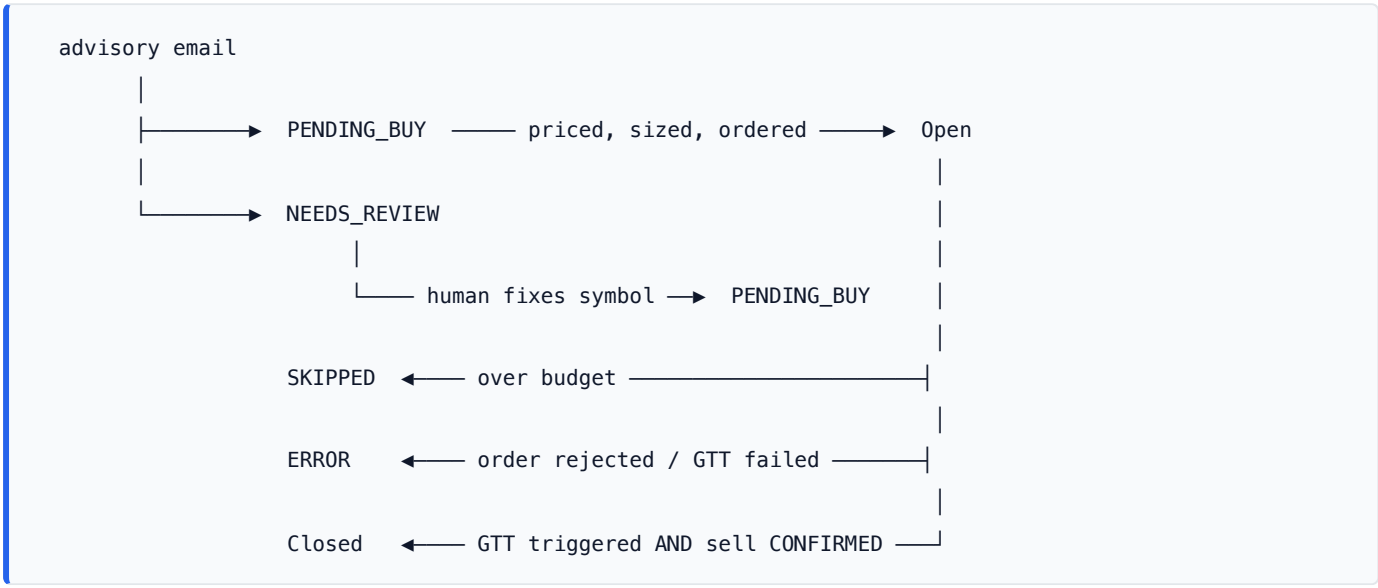
Each buy is capped by two independent rules, both read from the database:

| Rule | Meaning |
|---|---|
| **Category allocation** | Each advisory section (Big Gems, Little Gems, …) gets a percentage of total portfolio budget. |
| **Per-stock cap** | A single stock may not exceed a percentage of total budget, by market-cap class: Large 10%, Mid 6%, Small 4%, Micro 2%. |

The per-stock cap is **per individual stock**, not a shared pool across all stocks of that cap class. This distinction was originally implemented wrongly — as a shared bucket — which silently skipped legitimate trades once a few small-caps had been bought. Market-cap class comes from AMFI's official half-yearly categorisation, held in `STOCK_CAP_CLASSIFICATION`.

Budget is recycled when a position closes, so the ledger reflects capital actually at risk rather than cumulative spend.

### 2.3 Trade lifecycle




`Closed` requires a **confirmed fill**, not merely a triggered GTT. A GTT can leave the active state three ways: it triggered and the sell filled (a real close), it triggered but the day-validity sell order expired unfilled, or it was cancelled. Only the first closes the trade; the other two get a fresh GTT at the same target, tagged with a retry counter, so a position is never left silently unprotected.

### 2.4 The dashboard

A Streamlit application at `https://raramdas-stockbot.duckdns.org`, usable from a phone via Add to Home Screen. Login is a password with an encrypted 30-day "remember me" cookie.

| Page | Purpose |
|---|---|
| Overview | Budget deployed, live holdings, current value |
| Category Drill-Down | Holdings and P&L within one advisory section |
| Performance | Realised/unrealised P&L, cumulative chart, win rate |
| Set Targets | Edit target price / timeframe per open trade |
| Trades Explorer | Filterable full trade history, CSV export |
| Recommendations | Every tip ever seen, bought or not |
| **Conviction** | Evidence-based score per position, with full working |
| Needs Review | Fix a bad symbol, preview, then confirm a real buy |
| Open Orders | Live Kite orders and GTTs |
| Classification | AMFI market-cap lookup |
| Settings & Edits | Budget, allocations, manual trade close |

## 3. Technical view

### 3.1 Topology


Two deployment directories are kept in sync from one Git repository: `/home/ubuntu/stock_bot_v4` (cron jobs) and `/home/ubuntu/stockbot` (dashboard).

### 3.2 Egress split — the load-bearing constraint

`sptulsian.com` sits behind CloudFront, whose WAF rejects datacenter IP ranges **as a class**. The VM received HTTP 403 with CloudFront error headers on every path, including unauthenticated ones — the origin never saw the request. This is not rate limiting and not bot detection: no user-agent, header set, cookie jar or request pacing changes the outcome, because the discriminator is the source address.

Cloudflare WARP, run in **proxy mode**, egresses from consumer-registered addresses that the WAF serves normally. `warp-svc` exposes SOCKS5 on `127.0.0.1:40000`.

The critical property is **scope**. Zerodha's API is bound to this VM's registered static IP; a system-wide VPN or firewall redirect would move order placement onto Cloudflare's egress and break that binding. So the proxy is attached to exactly one object — the scraper's `requests.Session` — and nothing else:

```
# spt_scraper.build_session()
proxy = os.environ.get('SPTULSIAN_PROXY', '')   # socks5h://127.0.0.1:40000
if proxy:
    session.proxies = {'http': proxy, 'https': proxy}
```

Two details carry real weight:

- The variable is named `SPTULSIAN_PROXY`, **not** `HTTP_PROXY` / `HTTPS_PROXY`. `requests` honours the standard names automatically for every module in the process, which would silently route broker traffic through the proxy.
- The scheme is `socks5h`, not `socks5`. The trailing `h` resolves DNS at the proxy, so hostname lookups also traverse Cloudflare rather than leaking from the blocked address.

**The one check that matters.** If line 1 ever returns a Cloudflare address, the proxy has escaped its scope and broker traffic is at risk:

```
curl -s ifconfig.me                                  # → 140.245.226.35
curl -s --socks5-hostname 127.0.0.1:40000 ifconfig.me  # → a Cloudflare address
```

| Traffic | Egress | Why |
|---|---|---|
| Zerodha orders, holdings, GTTs | Static IP | Broker IP binding — must never move |
| Gmail IMAP, Oracle | Static IP | No reason to proxy |
| sptulsian.com, Yahoo, NSE | WARP SOCKS5 | The only traffic that is blocked |

### 3.3 Advisory portal scraping

The portal serves its sections through **two different backend controllers**, which requires two parsing strategies:

- **JSON** — the page wires up a per-section endpoint (`/getMWActiveData` for Little Gems, `/getFCActiveData` for Big Gems). The endpoint name is read off each page rather than hardcoded: calling one section's endpoint for another returns HTTP 500, which is easily misdiagnosed as a portal outage.
- **HTML** — the remaining sections ship no data endpoint and render calls directly into the page markup. Row markup is not positionally stable (the same table emits 11- and 13-cell rows, carrying both desktop and mobile cells), so fields are read from row text by content rather than column index.

Three correctness rules govern what is done with a scraped call:

**Login is verified on substance, not markers.** Checking for a "logout" link passes while signed out — the public teaser page contains that string in its navigation. A session is accepted only if it yields parsed call rows, which no signed-out response can produce.

**Archived is not the same as closed.** Little Gems and Big Gems carry no active list on this subscription, so even same-day calls appear under "Archives". Closure is determined by an actual exit remark (`Target met…`), not by which table a row sits in. Treating archived as closed would freeze out 82 of 86 open positions.

**Calls are matched within their own section.** The same stock is called in several sections with different targets and horizons — a Multibagger call may target 2027 at a far higher price than a 3-month Big Gems call on the same stock. Applying the wrong section's target arms a GTT that never triggers and parks the capital.

Regular Income Bluechips is excluded entirely: it is a covered-call section whose "Target" is a total position value (e.g. ₹7,80,000), not a per-share price.

### 3.4 Conviction engine

Answers a question the advisory does not: *is this call well-supported by public evidence?* Four weighted layers over a 100-point composite.

| Layer | Points | Checks |
|---|---|---|
| Fundamentals | 40 | Piotroski F-Score, Altman Z''-EM, Beneish M-Score, ROE, debt/equity, FCF yield, revenue growth |
| Consensus | 25 | Analyst rating, target upside, coverage breadth, advisory target sanity |
| Technical | 20 | 50/200-DMA alignment, RSI overbought guard, 52-week exhaustion, volume confirmation, liquidity |
| Governance | 15 | NSE ASM/GSM surveillance, promoter holding, institutional holding, results event risk |

Sources: Yahoo Finance (fundamentals, consensus, price history), NSE (surveillance). Both are fetched through the WARP proxy and both fail soft.

Three departures from the conventional design, each driven by what this portfolio actually contains:

**Missing data renormalises; it does not derate.** The portfolio is small- and micro-cap heavy, and 13 of 37 held symbols have zero analyst coverage. Deducting points for absent coverage would systematically penalise the entire Little Gems category — encoding "*we know less*" as "*this is worse*". Instead the score reports **quality** on what was measurable, and a separate **evidence** figure reports how much of the 100-point frame could be assessed. Layers enter the composite weighted by coverage, so a layer resting on one surviving check carries proportionally less weight rather than extrapolating that check across its whole budget.

`UNKNOWN` (tried and failed) reduces evidence. `N/A` (the metric never applied) does not.

**Financials skip the accrual models.** Piotroski, Altman and Beneish assume an operating balance sheet; current ratio, gross margin and asset turnover do not transfer to a lender. Financial Services names mark those checks N/A and renormalise rather than producing confident nonsense.

**Gates are separated from warnings.** A gate forces recommend-reject; a warning is surfaced and leaves the verdict to the score. Tuned against the live portfolio, where over-eager gates produced obvious false positives:

- **Beneish is a warning, never a gate.** Its sales-growth term over-flags fast growers; on this portfolio it flagged BEL, CG Power, Solar Industries and Mazagon Dock. A gate that rejects those trains the reader to ignore the tool.
- **Altman gates only below 3.0**, not at the 4.15 grey line. Vodafone Idea reads 0.57 and is unambiguous; capital-intensive manufacturers sit under 4.15 structurally.
- **Liquidity and GSM surveillance remain gates.** ₹0.00 crore/day of traded value means a GTT may never fill, regardless of how good the company is.

Below the evidence floor (40 of 100 points) no composite is published at all — a delisted symbol scoring "100/100" off one surviving governance check is worse than reporting nothing.

**The engine is display-only.** `main_conviction.py` writes to `CONVICTION_SCORES` and nothing else. It cannot size a position or stop a buy.

That caution was warranted. Sizing was briefly wired to the score on 2026-08-25 (>85 → ₹25,000, 75–85 → ₹10,000, below 75 not bought) and reverted the next day: `backtest_conviction.py`, rebuilding scores point-in-time and measuring excess return against NIFTY, found **no detectable relationship** — symbol-level Spearman −0.127 across 37 symbols (t = −0.76). The band that would have taken most of the capital performed worst. Two months and 12 realised closes prove nothing either way, which is the point: there was no evidence for a rule that was spending money.

*Validation:* backfilled across all 86 open positions, it independently flagged both trades already known to be mistakes — `ICDSLTD` (illiquid, ₹0.00 crore/day) and `NIFTY INFRA` (an index, insufficient evidence) — without being told about either.

### 3.5 The lite engine — what new recommendations actually get

`lib/conviction_lite.py` is the default (`main_conviction.py --engine lite`). Four components, one network call per symbol, no missing-filing holes:

| Component | Points | Why it is here |
|---|---|---|
| Momentum 12-1 | 35 | Most robustly documented equity factor globally and in India; the direction the per-check attribution pointed at |
| Trend alignment | 25 | The one technical check that correlated positively (rho +0.23) |
| Upside to advisory target | 25 | Already in the ledger, free, and it is the advisory's own stated edge |
| Liquidity | 15 | Not alpha — the constraint that decides whether a position can be exited at all |

**Its shape came from decomposing the full engine.** Correlating each individual check against realised excess return at symbol level exposed a contradiction inside the technical layer:

| Check | rho vs excess return |
|---|---|
| Overbought guard (RSI) | −0.348 |
| Trend alignment | +0.228 |
| 52-week exhaustion guard | −0.220 |
| Volume confirmation | +0.138 |
| Piotroski F-Score | −0.072 |
| Liquidity | +0.029 |
| Altman Z''-EM | −0.017 |

Trend alignment rewarded strength; the two guards penalised the same underlying property. They partly cancelled, which is a large part of why the composite landed near zero.

**So the guards became flags, not points.** "Don't chase an extended move" is a risk observation about *entry timing*. Encoding it as negative score marked down exactly the names that outperformed in that window. RSI, proximity to the 52-week high, realised volatility and "price already above target" are all surfaced on the dashboard and move no number. Liquidity is the sole hard gate, because an exit that cannot happen is a different risk in kind.

**Missing data still renormalises.** A recommendation with no scraped target yet drops the upside component and scores out of the remaining 75, reported as `evidence_pct = 75`. A missing target must not read as "no upside".

**This is a hypothesis, not a finding.** The attribution rests on 37 symbols over two months in one market regime, and at that size no single check clears a sensible noise threshold — the strongest is one of seven tests. The lite engine is therefore display-only on exactly the same terms as the full one. Scores from the two engines are stored with a `model` column and must never be pooled in a backtest.

### 3.6 Component reference

| Module | Role |
|---|---|
| `main_recommend.py` | **Phase 1** — resolve symbols, scrape targets, no orders |
| `main.py` | **Phase 2** — price, size, buy |
| `main_gtt_oracle.py` | **Phase 3** — place GTTs, confirm fills, close trades |
| `main_conviction.py` | Score today's recommendations (display only) |
| `spt_watchdog.py` | Liveness alarm for the scraper |
| `spt_capture.py` | Manual review-then-save of scraped targets |
| `lib/config.py` | Env/credentials, `KITE_ACCOUNT` switch (NEW/OLD), logging, IST |
| `lib/email_reader.py` | Gmail IMAP; parses the advisory email into tips |
| `lib/kite_client.py` | Kite login (TOTP), symbol resolution, order and GTT placement |
| `lib/order_status.py` | Order status lookup and sell-order reconciliation |
| `lib/budget_manager.py` | Budget policy, all `TRADES` reads/writes |
| `lib/spt_scraper.py` | Portal login, both parsing strategies, liveness watermark |
| `lib/conviction.py` | Four-layer fundamentals engine ( `--engine full` ) |
| `lib/conviction_lite.py` | Momentum/trend/upside/liquidity engine (default) |
| `lib/sheet_logger.py` | Google Sheets mirror (legacy, still written) |
| `dashboard/app.py` | Streamlit UI, all pages |
| `dashboard/db.py` | Dashboard's Oracle data access |
| `dashboard/kite_data.py` | Multi-account Kite sync, retry-buy preview/confirm |
| `dashboard/capture_api.py` | Authenticated endpoint for `spt_capture.py` |
| `dashboard/theme.py` | CSS design system, table and KPI rendering |
| `archive/` | Superseded and one-off scripts; nothing scheduled |
| `bot/` | Multi-tenant variant; keeps its own module copies, not scheduled |

Entrypoints sit at the repository root because cron invokes them by bare filename; everything they share lives in `lib/`. `bot/` deliberately does not import from `lib/` — it is a self-contained multi-tenant tree.

### 3.7 Data model

Oracle Autonomous Database, wallet authentication.

| Table | Contents |
|---|---|
| `TRADES` | The ledger. One row per recommendation, whatever its outcome. |
| `PORTFOLIO_BUDGET` | Total capital under management |
| `CATEGORY_ALLOCATION` | Per-section allocation % and per-cap-class caps |
| `STOCK_CAP_CLASSIFICATION` | AMFI market-cap categorisation |
| `CONVICTION_SCORES` | Score history; full working stored as JSON |
| `KITE_HOLDINGS_SNAPSHOT` | Holdings per account ( `account_label` NEW/OLD) |
| `KITE_GTT_SNAPSHOT` | Live GTTs per account |
| `KITE_ORDERS_SNAPSHOT` | Recent orders per account |
| `KITE_SYNC_LOG` | Per-account last-synced timestamps |

`TRADES` is the single source of truth. Every recommendation writes a row regardless of outcome, which is what makes the Recommendations page a complete record rather than a survivor-biased one.

### 3.8 Schedule

The VM clock is **UTC**; IST is UTC+5:30.

| Job | Cron (UTC) | IST | Purpose |
|---|---|---|---|
| `main_recommend.py` | `0 4 * * 1-5` | 09:30 | Phase 1 |
| `main_conviction.py` | `45 4 * * 1-5` | 10:15 | Scoring, lite engine (display only) |
| `spt_watchdog.py` | `15 5 * * 1-5` | 10:45 | Liveness alarm |
| `main.py` | `30 5 * * 1-5` | 11:00 | Phase 2 |
| `main_gtt_oracle.py` | `30 10 * * 1-5` | 16:00 | Phase 3 |
| DuckDNS refresh | `*/5 * * * *` | — | Dynamic DNS |

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## 4. Failure handling

### 4.1 The silence problem

A dead scraper and a quiet trading day look identical from outside. The buy flow deliberately continues when a scrape fails — recording the recommendation and resolving its symbol matter more than the target, which can be filled in later — so nothing in the pipeline raises. Without a dedicated check, the system could run blind for days while every log line still read "complete".

The fix is a **liveness watermark**: a timestamp written only after a section was genuinely fetched *and* parsed, never on a mere attempt. It is a signal a failed fetch cannot fake. Judging freshness from the newest row in `TRADES` was rejected — a genuinely quiet week would then raise a false alarm.

`spt_watchdog.py` applies two independent rules and emails on either:

| Rule | Condition | Rationale |
|---|---|---|
| Absolute | Watermark older than 30 h | Backstop for outages spanning a day |
| Trading-day | Past 10:30 IST on a weekday, watermark must carry today's date | The absolute rule alone would not fire until the next evening — after a whole session had run blind |

The scrape runs *before* the no-tips early return, making the watermark a true daily heartbeat. Otherwise a market holiday — a weekday with no tips — would be indistinguishable from an outage and would cry wolf every time.

### 4.2 Diagnostic matrix

| Symptom | Likely cause | Check |
|---|---|---|
| Connection refused to `127.0.0.1:40000` | WARP down | `sudo systemctl status warp-svc` ; `warp-cli status` |
| HTTP 403, CloudFront headers | Egress blocked or proxy bypassed | Compare direct vs proxied `ifconfig.me` |
| HTTP 500 from a section endpoint | Wrong controller for that section | Endpoint is read per page — confirm the page still declares it |
| Pages fetch, zero rows | Portal markup changed | `python3 spt_capture.py --dry-run` |
| Login "succeeds", no data | Credentials changed | Substance check should catch it; verify `SPT_USERNAME` / `SPT_PASSWORD` |
| TCP connects but **nothing responds** on :22 and :443 | Global OOM — userspace wedged, kernel still answering | OCI console → force reboot, then `journalctl -b -1 \| grep -i oom` |
| `warp-svc` restarting repeatedly | Memory leak hitting its cgroup cap | `systemctl show warp-svc -p MemoryCurrent` ; this is the cap working |

### 4.3 Safety invariants

These hold by construction and should be preserved by any future change:

1. Broker traffic egresses from the registered static IP. Always.
2. A symbol that does not resolve cleanly is never bought — it becomes `NEEDS_REVIEW` and waits for a human.
3. A trade is `Closed` only on a **confirmed** sell fill, never on a GTT status alone.
4. Conviction scores cannot size, gate, or block an order. They did for one day; the first backtest found no relationship between score and outcome, so it was reverted. See `backtest_conviction.py` and §3.4.
5. Manual buy paths (Needs Review, `spt_capture`) always preview before they commit, and the confirm step is the only thing that spends money.

### 4.4 Resource containment

The VM has **956 MB of RAM**. That is the binding constraint on everything here, and it is why the box went down on 2026-08-26.

`warp-svc` leaks. The kernel OOM-killed it on 18, 20, 22 and 24 August — roughly every two days, at ~250 MB resident each time — and on the 26th it grew fast enough to wedge the machine before the OOM killer resolved it.

The failure mode is worth understanding, because the symptom is misleading. With no cgroup limit, `warp-svc`'s growth triggers a **global** OOM, and the kernel then picks victims anywhere in the system. It took out `sshd` and Caddy. From outside, both ports still completed a TCP handshake — the kernel was healthy — but no daemon ever replied. Two unrelated services accepting connections and answering nothing is the signature of userspace starvation, not of a network or application fault. The disk, the obvious first suspect, was at 22%.

Three changes contain it:

| Change | Why |
|---|---|
| `MemoryMax=300M`, `MemoryHigh=220M`, `Restart=always` on `warp-svc` | Converts a global OOM into a local one. systemd kills only that unit and restarts it in 5s; the kernel never gets to choose `sshd`. Steady state is ~80 MB. |
| 2 GB swapfile, `vm.swappiness=10` | There was none. Swap is an emergency cushion here, not a working tier — hence the low swappiness. |
| `SystemMaxUse=200M` on journald | `warp-svc` dumps full metrics structures line by line when it degrades; journals had reached 632 MB, uncapped. |

The proxy is needed for about a minute a day, so a restart is invisible to the workload. If `warp-svc` ever begins restarting frequently, that is the cap doing its job — the leak is upstream in Cloudflare's daemon, not in this system.

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## 5. Operations

### 5.1 Deploy

```
# local
git add -A && git commit && git push

# VM — both directories track the same repo
ssh -i ~/.ssh/kite_key ubuntu@140.245.226.35 '
  cd /home/ubuntu/stock_bot_v4 && git pull -q
  cd /home/ubuntu/stockbot     && git pull -q
  sudo systemctl restart stockbot-dashboard'
```

### 5.2 Health check

```
ssh -i ~/.ssh/kite_key ubuntu@140.245.226.35 '
  curl -s ifconfig.me; echo                                    # must be 140.245.226.35
  curl -s --socks5-hostname 127.0.0.1:40000 ifconfig.me; echo  # must be Cloudflare
  sudo systemctl is-active warp-svc stockbot-dashboard stockbot-capture-api
  free -m                                                      # swap must be non-zero
  systemctl show warp-svc -p MemoryCurrent                     # ~80MB; cap is 300MB
  cd /home/ubuntu/stock_bot_v4 && python3 spt_watchdog.py --check-only'
```

On a 956 MB VM the memory line is not incidental — see §4.4.

### 5.3 Testing convention

Anything that touches money or the ledger is exercised in an isolated directory with Oracle writes, Sheet writes, and order placement monkey-patched out, run against **real live data**, and inspected before the real run. Mocked-write test harnesses have caught, among others: the pooled-budget bug, the GTT close that recorded no sell price, and a `DataFrame.__bool__` call that silently nulled every financial statement in the conviction engine.

Automated trading agents do not place real trades on the operator's behalf during development. Commands that move money are handed to the operator to run.

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## 6. Known limitations

- **WARP is a single point of failure** for the advisory feed, and a consumer service with no availability guarantee. Mitigated, not eliminated: the email channel still delivers call headlines, the watchdog surfaces an outage within one trading morning, and `spt_capture.py` remains a manual break-glass path.
- **Yahoo Finance is an unofficial source** for fundamentals. Coverage is good for this portfolio (all 37 held symbols return 4–5 years of statements) but is neither contractual nor audited.
- **Conviction governance is partial.** Promoter *pledge* and holding *trend* need shareholding-pattern filings, which are not yet parsed; only point-in-time holding percentages are scored.
- **Analyst consensus is thin for micro caps** by nature. Handled by renormalisation, but it means the Consensus layer contributes little for much of Little Gems.
- **Single tenant in practice.** `TENANT_CONFIG` and the `bot/` tree carry multi-tenant scaffolding that is not currently scheduled.

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## 7. Note on the conviction layer

The conviction engine computes published metrics from public data and shows its working. It is decision support, not financial advice, and not a prediction. Every threshold in it is a convention rather than a fact, tuned against one portfolio. The judgement stays with the human reading the output — which is precisely why the layer

is display-only and why every score is shown alongside the checks that produced it.